

The Accuracy of Respiratory Rate Estimation Using Electrocardiography and Photoplethysmography

Dan Wu, Guan-Zheng Liu, Mico Yee Man Wong, Yuan-Ting Zhang, *Fellow, IEEE*

Abstract—Respiratory rate can often be extracted from electrocardiogram (ECG) or (photoplethysmogram) PPG by analyzing its' beat morphological feature or the variability of time interval. In this paper, the respiratory rates were derived from ECG and PPG signals by using two algorithms based on beat morphology and time interval, respectively. The four groups of estimation results were evaluated by comparing with the reference respiratory rate collected using a piezoelectric sensor. In the *in-situ* experiments, ECG, PPG and reference respiratory signals were collected from ten subjects. To evaluate the performances of each algorithm at different respiratory rates, the subjects were asked to breathe at the natural rhythm, at the controlled respiratory rates and hold the breath during the experiment. The results demonstrated the method based on ECG peak feature was more accurate and stable among these methods.

Key words: Respiratory rate, electrocardiogram, respiration, photoplethysmogram

I. INTRODUCTION

Electrocardiogram (ECG), photoplethysmogram (PPG) and respiratory rate are vital physiological parameters and signals to assess the health status of patients with common chronic diseases [1]. Respiratory rate is traditionally measured using airflow detection method. This obtrusive technique requires a thermal sensor to secure to the patients' bodies and results discomfort to the patients. There are also some indirect methods using transducers to measure changes in body volume, for instance, piezoelectric transducer [2] and transthoracic inductance [3], each with associated burdens of wear.

Recently, the respiratory information can be extracted from the continuous monitoring of ECG and PPG [4, 5]. This helps to reduce the number of sensors attached to the patients' bodies during the measurement. S. P. Arunachalam

et al. used a new algorithm for baseline wander noise removal to estimate the ECG-derived respiration (EDR). The algorithm yielded EDR signals with a respiratory rate of 4.4bpm (breath per minute) [6]. Justin Boyle *et al.* proposed an automatic detection algorithm to derive respiratory rate from ambulatory single-lead ECG. Mean error rates with the reference acquired by air flow were ± 4 bpm (all activities), ± 2 bpm (lying and sitting), and ± 1 bpm (overnight studies) [7]. His work included more conditions in people's normal life, but the errors were on the high side. S. Dash compared the respiratory rate estimated from ECG, PPG and piezoelectric pulse transducer signals using the time-frequency methods [8]. The results showed that complex demodulation approach was better in both accuracy and repeatability but it was too complicated to implement in the real time.

In our previous study, we detected ECG and PPG based on contactless techniques and suggested a reliable algorithm based on beat morphology to estimate respiratory rate [9, 10]. In this paper, we added another method based on the Respiratory Sinus Arrhythmia (RSA, the cyclic variation in heart rate that is associated with respiration). Meanwhile, we further examined the two algorithms on two signals and reported the accuracy of respiratory rate estimation under the controlled experiments. The two algorithms had little errors and were so simple and low computation that can be used in the real-time monitoring for the chronic diseases.

II. METHODOLOGY

Experiments had been conducted on 10 healthy subjects (25 ± 3 years, from 23 to 28 years). We captured ECG, PPG and respiratory signal using a commercial data acquisition unit (MP150 from BIOPAC systems Inc., with ECG100C, OXY100C, RSP100C modules). Electrodes were placed on the subject's left arm, right arm and right leg for ECG detection; meanwhile, a TSD123A SpO₂ transducer was connected to the subject's index finger for PPG detection. The reference respiratory signal was detected using a piezoelectric sensor TSD201 embedded in a chest belt.

A. Experimental Procedures

The experiment lasted for 20 minutes for each participant. The subject sat on a chair and breathed at normal rate for 5 minutes at the beginning of the experiment. In order to evaluate the algorithms during different breathing conditions, ten subjects were asked to breathe at appointed respiratory rates (8, 12, 16, 20, 24 bpm). Music metronome was used to notice the subject to control his/her breaths at these

Manuscript received July 15, 2010. This work was supported by the 973 Project Fund (2010CB732606), the Guangdong Innovation Team Fund in China, and in part by the Hong Kong Innovation and Technology Fund (ITF).

Dan Wu and Guan-Zheng Liu are with the Institute of Biomedical and Health Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences and Key Laboratory for Biomedical Informatics and Health Engineering, Chinese Academy of Sciences, Shenzhen, China.

Mico Yee Man Wong was with the Joint Research Centre for Biomedical Engineering, The Chinese University of Hong Kong, Shatin, Hong Kong.

Yuan-Ting Zhang is with the Institute of Biomedical and Health Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, and Key Laboratory for Biomedical Informatics and Health Engineering, Chinese Academy of Sciences, Shenzhen 518055, China. He is also with the Joint Research Centre for Biomedical Engineering, Department of Electronic Engineering, The Chinese University of Hong Kong, Shatin, Hong Kong (phone: 0086-0755-86392249; fax: 0086-0755-86392276; e-mail: ytzhang@ee.cuhk.edu.hk).

appointed rates. Experiment in each appointed respiratory rate lasted for 2 minutes. Additionally, the subject held his breath for one or two times during one minute. We recorded ECG, PPG and respiratory signal throughout the experimental period. The collected analog signals were sampled at 1000 Hz by an A-D converter. The experimental procedures were illustrated in Table I.

TABLE I

THE PROCEDURES AND INFORMATIONS OF THE EXPERIMENTS

Experiment.	No. of participants	Respiration pattern	Time (min)
1	10	Natural breath	5
2	10	Controlled respiratory rates (8,12,16,20,24bpm)	10
3	10	Hold the breath	1

B. Data Analysis

The collected signals were analyzed offline using Matlab®. It is well known that the respiration influences electrocardiographic measurements with respect to both the heart rate variability and the beat morphology [11]. Therefore, briefly, the two methods were developed based on the variabilities of beat morphology and time-interval. The data processing procedure was illustrated in Fig.1. We detected the peaks of ECG R-wave and PPG, and calculated the time-interval of the peaks. Then we will get the ECG R-wave / PPG peak series and ECG R-R / PPG peak-to-peak series. Afterwards, the above two series were interpolated and re-sampled. We used polynomial fitting technique to reduce the baseline shift of interpolated signals. A FIR band-pass filter (0.2 and 0.4 Hz cutoff frequencies, 10th-order) was applied to the interpolated signals extracted from the beat morphology and time-interval algorithms. The respiratory rate was predicted every ten seconds using 1-minute data segment. The reference respiratory rate was calculated using the same method.

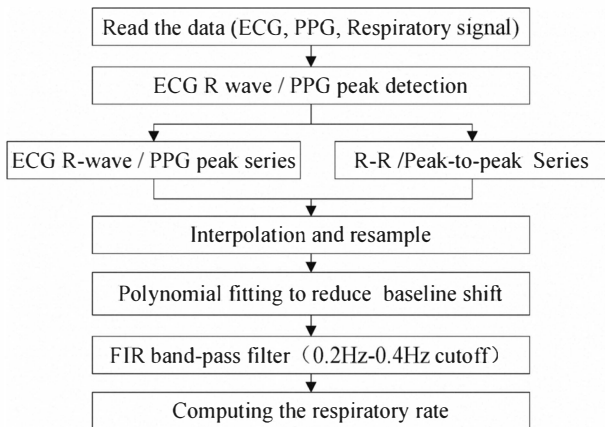


Fig. 1. Flow chat of the signal processing unit.

III. RESULTS

Fig.2 shows 1-minute data segment of four groups of derived respiratory signals and reference respiratory signal at different conditions for one subject. It was obvious that ECG/PPG derived respiratory signal had clear cycles and strong correlation with the reference signal. Only the

respiratory signal derived using PPG peak method had the baseline shift (see Fig.2 (a)). Additionally, for the peak detection had some mistakes, the extracted respiratory signal included some artifacts which would influence the monitoring accuracy, as shown in Fig.2 (a). From Fig.2 (c), we concluded that even when the subjects held the breath, the algorithms were also applicable.

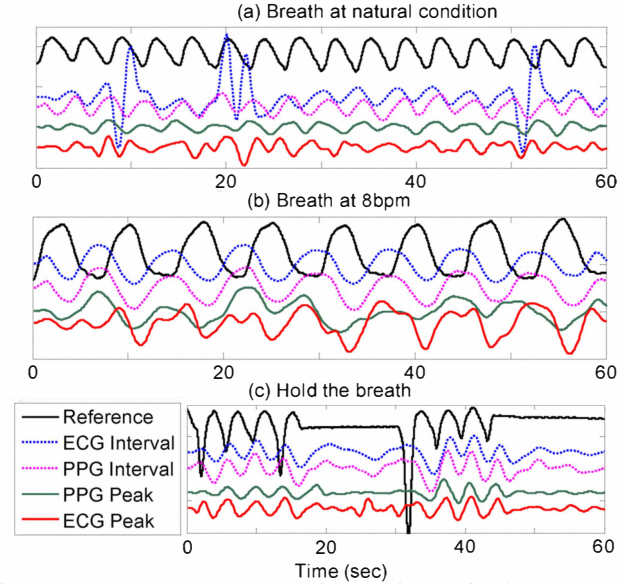


Fig.2. A segment of four groups of derived respiratory signals and reference respiratory signal at different conditions in a typical subject.

Table II presents that the difference of predicted respiratory rate for each individual under natural breathing condition. The difference of breathing cycles per minute was defined by Mean $\pm$ Standard Deviation (SD). It could be seen that the Mean and SD of absolute difference were 0.85 ± 0.53 and 1.64 ± 1.24 for ECG- and PPG-peak method. When respiratory rate was predicted using time-interval algorithm, the Mean and SD of absolute difference were 1.45 ± 1.29 and 1.46 ± 1.12 for ECG- and PPG-interval method, respectively. Thus, ECG-derived respiratory rate was more accurate than PPG-derived respiratory rate, especially the ECG peak method was more accurate and stable.

TABLE II

THE DIFFERENCE BETWEEN REFERENCE AND PREDICTED RESPIRATORY RATE IN EACH INDIVIDUAL AND OVERALL SUBJECTS

Subject	Beat morphology algorithm		Time-interval algorithm	
	ECG Peak	PPG Peak	ECG Interval	PPG Interval
1	0.48	0.72	0.88	0.36
2	0.64	4.84	0.92	1.96
3	0.72	1.64	1.04	2.16
4	0.88	0.76	0.76	1.08
5	1.14	1.71	3.86	3.86
6	0.71	2.29	3.86	2.29
7	0.64	1.16	0.48	0.28
8	0.6	0.64	0.56	0.44
9	0.4	1.64	0.88	1.32
10	2.24	0.99	1.24	0.84
Total Mean	0.85	1.64	1.45	1.46
Total SD	0.53	1.24	1.29	1.12

Table III shows the errors (Mean $\pm$ SD) between references and predicted respiratory rates at different breathing conditions for all individuals. The error was computed by equation (1).

$$\text{Error} = \frac{|\text{derived} - \text{reference}|}{\text{reference}} \quad (1)$$

Where derived meant respiration rate derived from ECG/PPG; reference meant the reference respiration rate derived from the piezoelectric sensor. Total error described the mean and SD of each method under all of the different breathing condition.

For all the subjects, PPG interval method had the best performance with the least difference. Meanwhile, PPG peak method performed worst with the largest difference.

TABLE III
THE ERRORS BETWEEN REFERENCES AND PREDICTED RESPIRATORY RATES
AT DIFFERENT CONDITIONS FOR ALL THE SUBJECTS

Conditions	Beat morphology algorithm		Time-interval algorithm	
	ECG Peak (%)	PPG Peak (%)	ECG Interval (%)	PPG Interval (%)
Breath at 8bpm	9.13 $\pm$ 3.75	11.63 $\pm$ 5.25	8.50 $\pm$ 3.13	6.75 $\pm$ 3.63
Breath at 12bpm	4.75 $\pm$ 3.25	8.67 $\pm$ 4.00	6.17 $\pm$ 6.42	5.58 $\pm$ 5.50
Breath at 16bpm	3.75 $\pm$ 2.25	6.25 $\pm$ 5.06	4.00 $\pm$ 3.13	3.75 $\pm$ 2.00
Breath at 20bpm	5.85 $\pm$ 4.20	5.15 $\pm$ 3.85	5.50 $\pm$ 3.80	4.65 $\pm$ 3.25
Breath at 24bpm	5.67 $\pm$ 4.88	8.20 $\pm$ 8.67	5.71 $\pm$ 5.25	4.88 $\pm$ 2.79
Total Error	5.83 $\pm$ 2.02	7.98 $\pm$ 2.49	5.98 $\pm$ 1.63	5.12 $\pm$ 1.12

Fig.3 indicates that all the errors for ECG/PPG derived respiratory methods were blow 10%. The PPG interval method with the error at 5.12% was the best performance and PPG peak method was the worst performance.

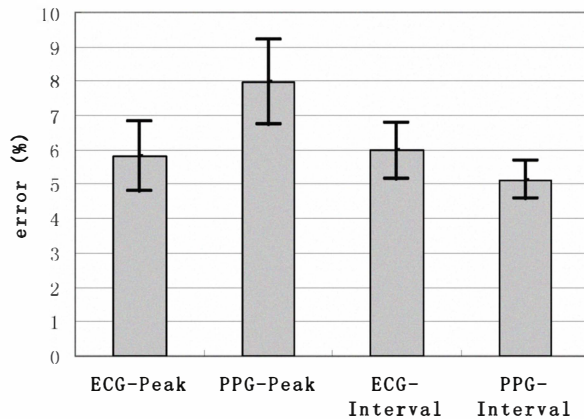


Fig.3. Total errors for different methods under different respiratory rates (8-24bpm) for all the subjects. Mean $\pm$ Standard Deviation (SD) was plotted.

IV. DISCUSSION AND CONCLUSION

In this study, we estimated respiratory rate from contact bio-signals using two simple and low computation cost algorithms. We further examined the algorithms in different experimental conditions.

The following conclusions were drawn from this study.

1) Under the natural breathing condition, compared with PPG signal, ECG signal was more easily influenced by respiration. Thus, ECG-derived respiratory rate showed smaller prediction difference than PPG-derived respiratory rate. The ECG peak method was the most accurate and PPG peak method performed the worst.

2) For the controlled respiratory rate tested in this study (8, 12, 16, 20, 24bpm), beat morphology and time-interval algorithms both had good performances with the error below 10%. The worst result also came from the PPG peak method. As the PPG signal was interfered by the subjects' motion, the peaks of the PPG signal would fluctuate irregularly along with the motion artifact.

3) During the subjects held their breath, the amplitude of derived respiratory signal decreased obviously. Thus, both of the two algorithms can be used in the apnea.

4) In general, the ECG peak method was the best method under different conditions.

In the future, we will do the further research on the effects that have influence on the accuracy of the derived respiratory rate. Meanwhile, the robust algorithms will be used to reduce these interferes. The data from cardiac and respiratory diseases should be included in the analysis.

REFERENCES

- [1] S. Park, S. Jayaraman, "Enhancing Quality of Life through Wearable Technology", *IEEE EMB Magazine*, May / June 2003, pp. 41-48.
- [2] M. H. Jones, R. Goubran, F. Knoefel. "Reliable Respiratory Rate Estimation from a Bed Pressure Array", *the 28th Ann Int Conf of IEEE-EMBS*, New York City, USA, pp.6410-6413, 2006.
- [3] D. Wu, L. Wang, Y. T. Zhang et al. "A Wearable Respiration Monitoring System Based on Digital Respiratory Inductive Plethysmography". *the 31st Ann Int Conf of IEEE-EMBS*, Minnesota, USA, 2009.
- [4] D. Cysarz, R. Zerm, H. Bettermann et al., "Comparison of Respiratory Rates Derived from Heart Rate Variability, ECG Amplitude, and Nasal/Oral Airflow", *Annals of Biomedical Engineering*, vol.36, no.12, pp.2085-2094, 2008.
- [5] E. M. Lee, N. H. Kim, N. T. Trang et al., "Respiratory Rate Detection Algorithms by Photoplethysmography Signal Processing", *the 30th Ann Int Conf of IEEE-EMBS*, British Columbia, Canada, pp.1140-1143, 2008.
- [6] S. P. Arunachalam, L. F. Brown, "Real-Time Estimation of the ECG-Derived Respiration (EDR) Signal using a New Algorithm for Baseline Wander Noise Removal", *the 31st Ann Int Conf of IEEE-EMBS*, Minneapolis, Minnesota, USA, pp.5681-5684, 2009.
- [7] J. Boyle, N. Bidargaddi, A. Sarela, M. Karunanithi, "Automatic Detection of Respiratory Rate From Ambulatory Single-Lead ECG", *IEEE trans. on Information Technology in Biomedical*, vol.13, no.6, pp.890-896, 2009.
- [8] S. Dash, K. H. Shelly, D. G. Silverman et al., "Estimation of Respiratory Rate From ECG, Photoplethysmogram, and Piezoelectric Pulse Transducer Signals: A Comparative Study of Time-Frequency Methods", *IEEE trans. on Biomedical Engineering*, vol. 57, no. 5, pp.1099-1107, 2010.
- [9] M. Y. M. Wong, H. K. Leung, E. P. MacPherson et al., "Contactless recording of photoplethysmogram on a sleeping bed", *in the Proc. of the 31st Ann Int Conf of IEEE-EMBS*, Minneapolis, USA, 2009.
- [10] D. Wu, M. Y. M. Wong, Y. T. Zhang, "Respiratory Rate Estimation Using Contactless Electrocardiography and Photoplethysmography", *the 6th conference of MDBS-BHE*, Shenzhen, China, 2009.
- [11] C. Orphanidou, O. Brain, J. Feldmar et al., "Spectral fusion for estimating respiratory rate from the ECG", *Proceedings of the 9th Int Conf of ITAB*, Larnaca, Cyprus, 2009.